WAJIDA MUSRAT TAMBOLI

Email LinkedIn Dharwad, Karnataka

Summary

AR/VR & XR Developer with expertise in building immersive experiences using Unity and Unreal Engine. Proficient in C# and C++, specializing in Mixed Reality applications for Meta Quest, including hand tracking, AR/MR interactions, and performance optimization. Experienced in 3D modeling, Digital Twin integrations, and industrial training simulations, with a strong passion for next-generation spatial computing.

Experience

Abhyudyaya Techno Solutions Pvt Ltd **02/2024 – Present**

Junior Software Developer **Hubli, KA**

Immersive Technology (AR, VR, MR & XR) - Unity 3D

- Developed immersive Mixed Reality applications for Meta Quest 3 using Unity, XR Interaction Toolkit, & OpenXR.
- Integrated hand tracking, passthrough, and spatial anchors to deliver intuitive and context-aware MR experiences.
- Designed and implemented interactive 3D user interfaces for industrial training and educational simulations.
- Optimized applications for standalone XR headsets, achieving stable frame rates and reduced latency.
- Built custom interaction systems using raycasting, grab interactors,socket-based mechanics, & physics-driven logic.
- Implemented AR marker tracking and plane detection using AR Foundation for hybrid AR/MR workflows.
- Developed modular and scalable C# scripts for object interactions, state management, and environment logic.
- Led Digital Twin integration, mapping real-time sensor data to virtual assets for live system visualization.
- Collaborated in agile development teams to debug, test, and deploy feature-rich XR modules.
- Contributed to UI/UX decisions, improving usability, accessibility, and user immersion,
- Prototyped and deployed XR training simulations for Meta Quest 3 and companion mobile applications.
- Explored and optimized rendering using URP, Shader Graph, and OpenXR pipelines.
- Documented development workflows and created reusable templates to accelerate future XR projects.

Projects

VR Simulation – Lathe Machine Operation

- Built an immersive VR training system for lathe machine operation using Unity and Meta Quest 3
- Simulated spindle rotation, tool movement, cutting depth, feed rate, and emergency stop logic in C#
- Integrated hand tracking for tool handling, safety checks, and measurements
- Reduced dependency on physical machines by 55% through safe, repeatable VR training
- Used XR Interaction Toolkit and Shader Graph for realism and feedback

MR Smart Agriculture – IoT-Based Digital Twin

- Developed a Mixed Reality Digital Twin for smart agriculture visualization
- Displayed live IoT sensor data (soil moisture, temperature, NPK, pH, EC)
- Built interactive MR dashboards and simulated irrigation & fogger control
- Promoted precision farming and sustainable agriculture using MR

VR Bricklaying Training Simulation

- Developed a VR module for teaching brick placement and alignment
- Implemented hand-based brick placement and mortar application logic
- Added tutorial guidance and scoring system for skill assessment
- Targeted vocational training institutes for safe skill development

MR Home Automation – IoT Digital Twin

- Created a Mixed Reality smart home application using Unity and Meta Quest 3
- Enabled real-time control of lights, fans, and appliances via virtual UI
- Integrated passthrough and scene understanding for physical–digital blending
- Demonstrated MR use-cases for energy-efficient smart living

VR Simulation – Plumbing Installation & Maintenance

- Designed a hands-on VR training module for plumbing tasks
- Simulated water flow, pressure, leaks, and tool interactions using C#
- Enabled step-by-step scenarios like pipe cutting, sealing, and testing
- Reduced material waste and rework by 50%
- Used XR Interaction Toolkit, Shader Graph, and particle systems

AR Studio – Mobile App

- Built an AR application for industrial robot layout visualization using AR Foundation
- Enabled placement, scaling, and rotation of 3D robot models in real environments
- Supported factory layout validation and spatial planning
- Reduced physical prototyping cost and improved decision-making speed

VR Simulation – Factory Robotic Arm Operation

- Created a smart factory VR environment with robotic arms and conveyors
- Programmed pick-and-place, welding, and sorting using inverse kinematics
- Integrated hand tracking and UI controls for task programming
- Provided real-time feedback on joint angles, speed, and safety zones
- Improved operator readiness and safety compliance by 65%
- Used XR Interaction Toolkit, Timeline, and Animation Rigging

AR-Based Speech-to-Text Quiz Application

- Created an AR quiz system with real-time voice-to-text interaction
- Enabled users to answer quizzes verbally, with spoken responses transcribed and evaluated instantly
- Integrated AR interfaces to display questions and scores in the physical environment
- Implemented voice recognition and text processing for accurate response matching
- Designed an intuitive hands-free learning experience, improving engagement and accessibility
- Targeted education and training use-cases, blending AR visualization with voice-driven interaction

Skills

C# (Unity) | AR/VR Frameworks | ARKit and ARCore | 3D Modelling and Animation | UI/UX design | Spatial Computing | Gesture and Motion Tracking | Cross-Platform Development | Hardware Integration | AWS Cloud| C++ | C# | IBM Cloud IoT | Java | Python | Unity | Augmented Reality | Virtual Reality | Mixed Reality| Vuforia |

Education

KLS Vishwanathrao Deshpande Institute of Technology Haliyal — **Haliyal, KA**
Bachelor of Engineering in Electronics & Communication — **June 2022**

JSS Shri Manjunatheswara IND.P.U Science and Commerce College Dharwad — **Dharwad, KA**
Higher Secondary in Science — **April 2018**

Pavan English Medium High School Dharwad — **Dharwad, KA**
Secondary — **April 2016**

Languages

English – Fluent **Kannada –Fluent** **Hindi - Native**